

Project report Hotel management system

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over view

Our project is the Hotel Management System and it consists of eight classes. The project starts by requesting a login from the user and verifying their registration. After that, a menu for managing clients opens. Then a list appears showing the rooms available for the user to choose from, after which the user shows the room prices and types. The user is then asked for the number of days he will spend in the room, then the user is asked for the number of rooms he wants to book, and after that the final price is shown to the user. Then a list of services available in the hotel opens, such as the Internet and other services. After that, the user selects the services he desires, and is shown a list of prices for the services he chose. Finally, the program ends

Class USERNAME :

The 'Username' class represents a user authentication and registration system.

It allows users to log in with an existing username and password or create a new account.

The class provides methods for welcoming users, validating their input, checking username availability, and adding user information to a text file. It also includes a 'main' method for user interaction.

Key Features:

- User authentication and registration functionality.
- Username and password validation.
- Age verification for certain content access.
- Storing user information in a text file.
- Console-based user interface for login and registration.

This class relies on external text files for user data, so ensure the presence of these files.

CLASS CUSTOMER:

"Customer" class in the Java programming language. A class is a blueprint or template for creating objects. It defines the properties (variables) and behaviors (methods) that objects of that class will have. In this case, the Customer class has variables like customerFirstName, customerLastName, customerNumber, customerEmail, customerAddress, and age. It also has getter and setter methods to access and modify these variables. By creating objects of the Customer class, you can store and manipulate customer data.

CLASS CUSTOMERFILEMANAGER:

class called "Customer File Manager" that deals with reading and writing customer data to a file. It has a method called "writeCustomer Data" that takes an ArrayList of Customer objects as a parameter and writes it to a file using ObjectOutputStream. It also has a method Halled "readCustomerData" that reads customer data from a file using ObjectInputStream and returns an ArrayList of Customer objects. There is also a method called "searchCustomers" that takes a search term as a parameter and returns an ArrayList of Customer objects that match the search term.

CLASS RoomService:

This class represents a hotel service that allows users to book rooms and view available rooms. It utilizes an enum called RoomType to define different types of rooms available in the hotel . The class provides methods for booking a room and displaying available rooms.

CLASS Hotel class :

The 'Hotel' class represents a hotel entity that offers a list of services to its guests.
It contains information about the hotel's name and a collection of services it provides.

CLASS Service class :

The 'Service' class represents a service that can be offered by a business or entity.
It contains information about the service's name and price.

CLASS 'ServiceSelector'

The 'ServiceSelector' class represents a service selector program that allows users
to choose from a list of available services and select the ones they want. It provides
an interactive console-based interface for users to select services and calculates
the total price for the chosen services

CLASS main :

is a simple customer management system. It allows users to add, search, modify, delete, and print customer data. The system is implemented using a few key components:

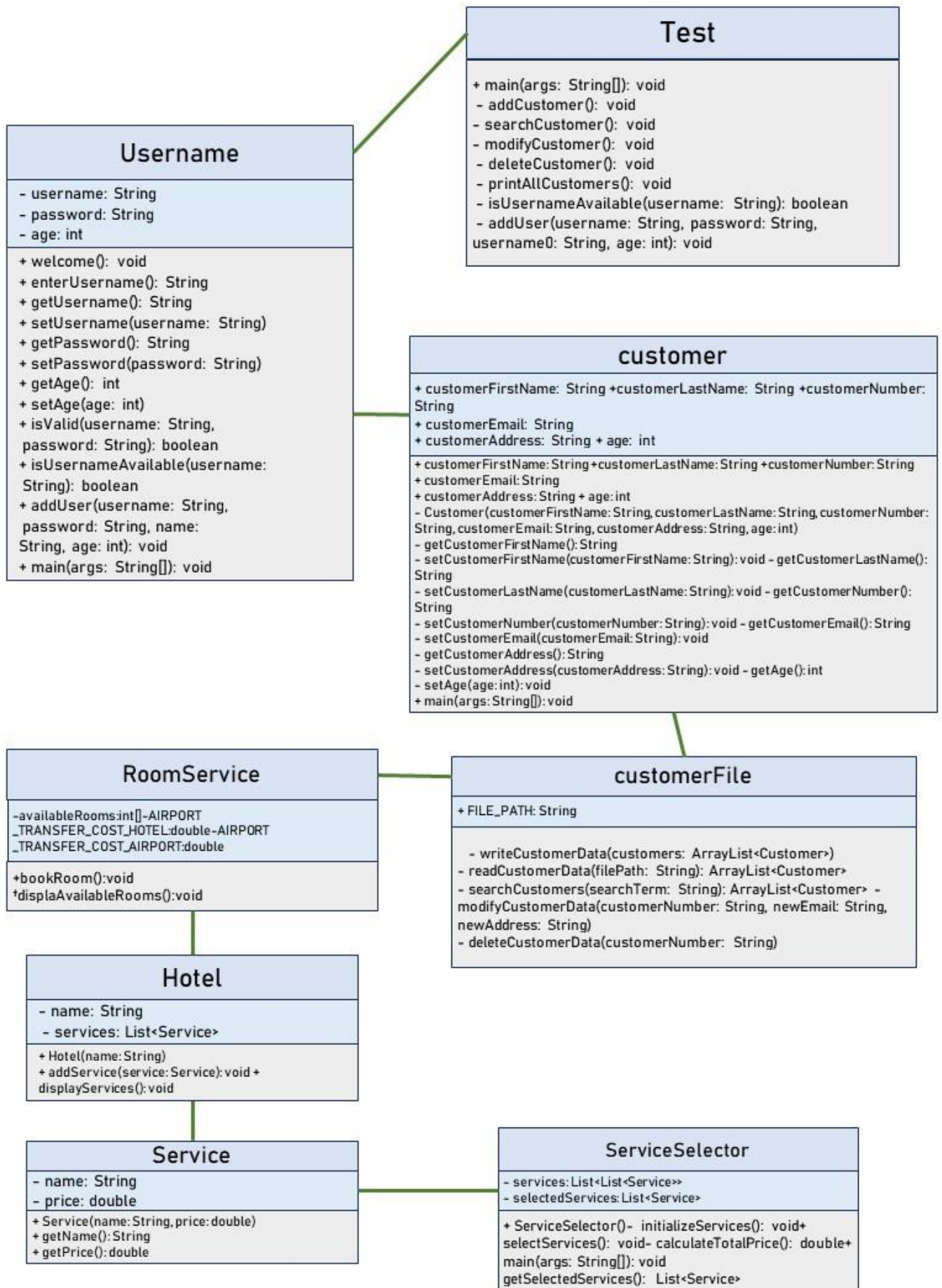
‘Customer class:’ This class represents a customer and stores their basic information, such as name, customer number, email, address, and age.

‘ArrayList:’ This data structure is used to store a list of customers.

‘CustomerFileManager class:’ This class provides methods for reading and writing customer data to a file.

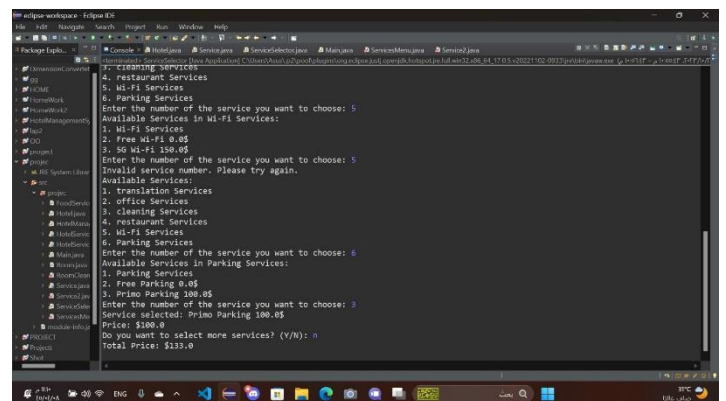
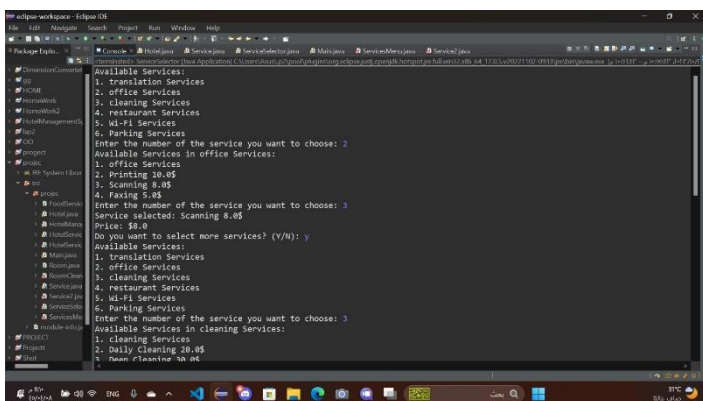
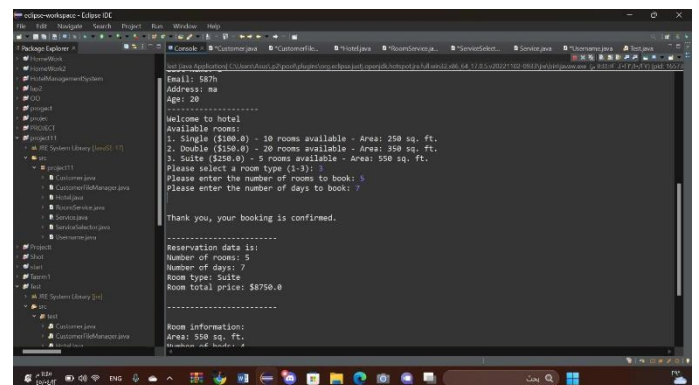
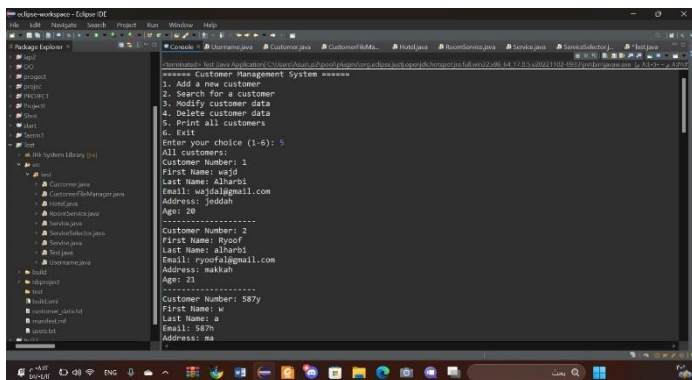
The main method of the Test class implements the main logic of the customer management system. It presents a menu of options to the user and allows them to perform the desired operation.:

UML diagram



Project testing

```
Window Help
Console x Username.java
<terminated> Username [Java Application] C:\Users\dxv80\p2\poolp...
Please choose an option:
1. Login
2. Create a new account
3. Exit
2
Create a new account selected.
Welcome to our Hotel
Enter your username:
amal
Enter your password:
12345678
Enter your age:
22
Welcome amal!
Account created successfully!
Please choose an option:
1. Login
2. Create a new account
3. Exit
1
Login selected.
Welcome to our Hotel
Enter your username:
amal
Enter your password:
12345678
Enter your age:
22
Welcome amal!
Welcome back, amal!
Please choose an option:
1. Login
2. Create a new account
3. Exit
3
4
```



Incoming errors

The screenshot displays the Eclipse IDE interface with a project named "aspire-workspace - Eclipse IDE". The Package Explorer on the left lists various Java packages and classes, including "build", "srcproject", "build.xml", "customer_data.txt", "Customer.java", "CustomerInfoManager.java", "Hotel.java", "HotelInfo.txt", "RoomService.java", "Service.java", "ServiceSelector.java", "Service.java", "User.java", and "UserList".

The main editor window shows the "Customer.java" file. The code defines a class with methods for logging in, creating accounts, and exiting. The console output at the bottom shows the following sequence of events:

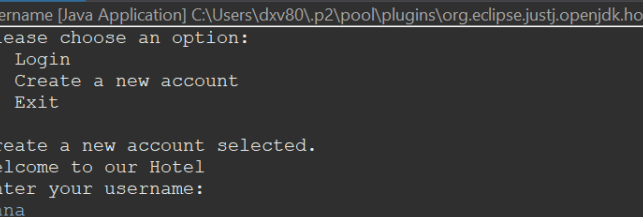
```

Welcome to Hotel Application
Please choose an option:
1. Login
2. Create a new account
3. Exit
1
Login selected.
Welcome to our Hotel.
Enter your username:
amal
Enter letters only.
Enter your username:
amal
Enter your password:
123erty
Password must be 8 characters long. Please try again.
Enter your password:
123ertyfg
Invalid password.
Enter your age:
17
Sorry, this content is not available for users under 17 years old or over 99 years old. Please try again.
Enter your age:
26
Welcome amal!
Invalid username or password Please Create a new account .
Please choose an option:
1. Login
2. Create a new account
3. Exit
2

```

The console also shows some system messages related to the JRE System Library.

```
Run Window Help
C:\Users\dev00\p2\pool\plaine\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64.j7.0.5.v20221102-0933\jre\bin\java.exe (-Xms128M -Xmx1T)
Console x Username.java
Username [java Application] C:\Users\dev00\p2\pool\plaine\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64.j7.0.5.v20221102-0933\jre\bin\java.exe (-Xms128M -Xmx1T)
3. Exit
1
Login selected.
Welcome to our Hotel
Enter your username:
aaaaa
Enter your password:
123456789
Enter your age:
17
Sorry, this content is not available for users under 17 years old or over 99 years old. Please try again.
Enter your age:
18
Welcome aaaaa!
Invalid username or password Please Create a new account .
Please choose an option:
1. Login
2. Create a new account
3. Exit
2
Create a new account selected.
Welcome to our Hotel
Enter your username:
aaaaa
Enter your password:
123456789
Enter your age:
18
Welcome aaaaa!
Account created successfully!
Please choose an option:
1. Login
2. Create a new account
3. Exit
```



The screenshot shows a Java application window titled "Console x Username.java". The application is running in the directory "C:\Users\dxv80\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full...". The console output shows the following sequence of events:

```
Username [Java Application] C:\Users\dxv80\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full...
Please choose an option:
1. Login
2. Create a new account
3. Exit
2
Create a new account selected.
Welcome to our Hotel
Enter your username:
dana
Enter your password:
12345678
Enter your age:
18
Welcome dana!
Username is already taken Please Choose another one.
Please choose an option:
1. Login
2. Create a new account
3. Exit
```